



How visualization can be an essential tool for exploring and communicating complicated information Transcript

Hello everyone, welcome to BITS Pilani Digital. Today we will continue our learning journey for the course Data Visualisation and Storytelling. We will be continuing our module Designing Visuals, Visual Best Practises.

Now we will look at how to visualise structured data and what are the best charts for the given purpose. Before we dwell into today's lesson, we want to understand the difference between structured and unstructured data. Structured data which is structured in form of matrices with clear distinction of variables, type, and the contents that are therein.

The very fact it is structured, you can understand the data set pretty easily or relatively, and it is amenable for computational in an easy format. For example, you have numerical data, the sales data of the company, where you know which customer is buying how much of goods through which channel and how much is the revenue generated. It's a small example where there are four variables and it might run into lakhs of those depending upon your customers, but all the data points are clearly known to you.

What kind of variable can come and sit. Now as opposed to this, you have unstructured data, which is generally text data in which there is no proper structure to the information that you are getting. For example, there is a customer complaint email or a customer review on Twitter.

So this can be wide ranging in terms of content, texture and format. So the data there, the information that is contained in that particular message, text message or in the email is not readily amenable for structuring into a particular type of variable or is not able to directly or we are not able to directly create some information out of that message. One has to read the message, then only you can understand what is there in the text message.

As opposed to this, in a structured data, once the variables are defined, you can easily start the computational exercises and therefore visualising structured data is relatively easier than visualising the unstructured data. Today, we will be restricting our attention to handling the structured data. Within the structured data, where we function in most of our business scenarios, as far as the analysis is concerned, we try to make the data more and more structured so that the computations and the visualisations analysis can be done in a coherent fashion.

Like in last class, we have discussed the different types of data. We have categorical data, ordinal data, and the binary data and numerical data. Let me get a little bit into the details.

Categorical data or the nominal data are the data that represent a particular category like gender or race. Then you have binary data, which is of two categories. You can

think them as a special category of categorical data, but they represent a phenomenon happening or not happening.

Generally, they are like mutually exclusive series. For example, whether a customer visiting your website makes a purchase or not, whether you book a ticket and you get it confirmed or not. So if one of the classes happens, it is not the other class doesn't happen.

So you cannot be on both cases. So binary variables are used to understand the behaviour or a particular pattern is being present or not, whether you are purchasing or not purchasing, whether you are getting a confirmed ticket or not, or whether you are in default, credit default, or you're paying your loans on time, you cannot be both. So this is a binary category.

Then you have the ordinal data, which we have discussed earlier, like the review ratings, which can be ranked from a given position 0 to 1 or 1 to 2, like that. Ordinal data, where you have ranking, like customer reviews on a scale of 1 to 5 or 1 to 10, you can then make comparisons on them. Then you have numerical continuous data, which can take a continuous value like revenue, growth, sales, or revenue, growth, sales, GDP, like this.

Whereas you have numerical discrete data, which pertain to number of cars sold, right, a number of houses that are there, there you cannot, though the data is numerical, you count them, you can do the calculations, they cannot be in fractions, they have to be whole units, you have 10 cars, you have 11 cars, but you cannot have 10.5 cars. So these distinctions are important while we deal with the structured data, though all of them fall into structured data, because you are able to clearly identify the categories, types of the data that is representing, we are able to clearly identify the categories and types of the information that the data is representing, and we can use them for building our visualisations. So again, before starting any visualisation exercise with structured data, please understand the variable types that you have.

Therefore, you can use the appropriate tools and the chart types to bring out the information in an efficient form. And what is the main objective of this exercise? The exercise is to use your visual to draw appropriate inferences or use your visual and make your audience draw appropriate inferences. We'll start with univariate analysis.

Let us say your data set has multiple variables. Univariate analysis is by the name, looking at a particular variable in the data set. You'll always try to analyse the data set from a particular variable.

For example, let us talk about the online retail sales data, which we'll be dealing in the experiential learning session also, where the data captures about the behaviour of customers who are visiting a particular retail commerce website and trying to make a purchase. We are looking at the number of times the customer is visiting, the type of customer, and whether he's doing a purchase or not, and what is the revenue or other

attributes. So the data comprises of about 10 to 12 attributes, and of which we are interested at any given point in time or a one attribute.

Such analysis is called a univariate analysis. So for now, we will be looking at whether the customer is making a purchase or not. The most important characteristic of the data set is to identify whether a customer coming to the website landing on the web page is making a purchase or not.

This is called the conversion rate. In this case, the conversion rate is 15.5%. How is this computed? There are 1908 sale transactions or purchase transactions in the combination of some 15,000 transactions put together, sorry, put together some 12,000 transactions, you get this 15%. The point here is, you know, the main number is 15%.

You can simply put the 15%. If you remember, simple text is your best friend. So somebody asked what is the key takeaway from the data set from a retail commerce standpoint, the conversion rate for this particular retail commerce company or website is 15.5. You can simply show that.

So you are now focussing on a single variable called conversion rate. And that is the answer. Then you can now look at whether the customer has bought or not, the number of people who have purchased and the number of people who have not purchased.

So you can simply take a bar chart. Remember bar chart is used for comparisons. What are you comparing? You are comparing customers who have purchased and who have not purchased.

So you have 10,442 customers who have not purchased despite visiting the website. And then you have 1908 customers who have purchased. So this is a highly imbalanced data with the buyers being very less in number as compared to the people who have not bought.

This becomes very important as you go for the AML based modelling. Imbalanced data sets might give you skewed results. It is important to first do an exploratory data analysis using univariate techniques or multivariate techniques as we will see in a couple of slides to understand the underlying data and do the appropriate corrections or transformations before you deploy your models.

Similarly, you can understand the composition of the customers. People have bought and people have not bought using a pie chart or a donut chart by simply saying that 15.5% is the conversion rate, which means 84.5% of the people who visited the website did not buy. So these kinds of information or comparisons can be done using univariate analysis and it is the first step in the exploratory analysis.

Now you can extend this univariate analysis by trying to understand the relationship of the main variable of interest with others. For example, are people buying more during a weekday or on a weekend? So you can simply get the data of weekend and

see whether they have bought on weekend or whether they are buying on the weekday. So weekend false, weekend true.

So whether they are engaging on the weekend or on the weekday. So actually the weekday visits are higher than the weekend visit. This is number of records customers visit as on weekday or weekend.

You can extend univariate analysis to understand the relationships between different type of attributes in the data or the variables in the data. So now the company is interested to understand whether the customers are visiting them on a weekend or on a weekday. So you can see that of the customers that are totally visiting the website, 9462 have visited on a weekday as compared to 2868 on a weekend.

So the company can plan the internet traffic management depending upon how many people are visiting. Another way to do is per day average and we can do all these kind of comparisons to get in some insight. Similarly, we can see the number of visitors as per the visitor type.

Returning visitor who is a repeat customer or a new visitor or some other person. Then you have breakup of volumes as per the regions. If your retail company or the website is present in different regions or your store is in different regions, you have sales from different regions, you can present and do a comparison.

So all this is univariate analysis. What is that you are trying to do? You are looking at a particular variable and see how is it distributed or how is it functioning with relation to other attributes in the data set. For example, we are seeing the regions and looking at the visitors.

Number of visitors is the variable, then regions is the parameter, then again number of visitors you have weekend and weekday, then you have number of visitors, then you have type of visitor. So you are keeping the one particular parameter in mind and trying to see how are the visits of the website. So you can use univariate analysis to understand the relationship between the relationship among the variables and then use them for building your insights.

Another important graph type in univariate distribution is histogram. This helps us to understand the underlying data distribution. Histogram is simply nothing but the frequency plot of a particular class interval.

For example, you can take how many customers have visited less than 50 times, then you have a large number of people and very few people visiting your website more than 50 times and even more fewer people visiting more than 100 times. So when you do a frequency distribution, the histogram, it clearly tells you what is the underlying data structure and that will give you an idea about where to focus to get the maximum information in the data set. Now we go to another tool called boxplot.

Boxplot is an extension of an histogram, but it tries to present the distribution using the quartiles. So the boxplot shows the 25th percentile, the 75th percentile and the median

and the whiskers are 1.5 times of the interquartile distance, which is nothing but the distance between upper quartile and lower quartile. If the distribution is perfectly normal, this is how it would look symmetrical on around the median with equal distance on 25th and the 75th percentile.

The problem here is generally the distributions are not normal and using boxplot you can see whether the data is due to the right or to the left, how distant is the median and whether there are any outliers. Outliers would be which are away from the upper whisker or lower whisker. So you can see the distribution of the data in a single snapshot and draw very insightful conclusions using boxplot.

So let us use boxplot. We just had the histogram of bounce rates. So bounce rates histogram is simply showing that you have very few bounce rates, then we have skewed distribution of bounce rate.

Now the same thing is presented in the boxplot. The boxplot shows the skewness in the data that there are very few bounce rates between 0.000 to 0.00 to 0.05 with very few outliers. The outliers are but the outlier values are very high.

There are few outliers but there are very outlier values are very high. So if you see in the histogram also you can see the outliers but it is very difficult for the eye to capture this. Whereas in a boxplot you can readily show the outliers and also the nature of distribution.

The same conclusions will come. Here also the bounce rate is very low for majority of transactions or the visits. Here the bounce rate also is very low it is closely packed to 0 near to 0 but then you have few outliers that are clearly shown which are easy to read as compared to a histogram.

Therefore boxplot is an improvement of histogram. So whenever you are trying to see the distribution look at the histogram also look at the boxplot so that you have clear insights on the data distribution. Now we move to the multivariate analysis.

Multivariate analysis the very name suggests that there are two or more variables that are involved. As opposed to univariate analysis where you focused only on one particular variable in multivariate analysis you are talking about two or more variables and their comparisons or relationships. We start with a stacked bar graph as compared to the normal bar graph used for univariate analysis.

In this case we are seeing the weekend sales and the weekday sales. For example we are asking the question do we have higher sales in the weekend than the weekday. By the numbers you are having naturally higher number of sales during the weekday as compared to the number of sales that are there in the weekend.

However when you try to take the proportion of sales in that are done in the weekday versus weekend you can simply understand that the sales that are done on the weekend are higher compared to the overall visits to the website as compared to the sales that are done during the weekdays. So this insight also can be brought by

comparing a stacked bar graph as a proportion of the sales. For example the first bar graph that we have used the stacked bar graph is an absolute scale where you are just plotting the number of visits and the differentiation.

So what are the two variables that we are seeing? We are seeing whether they have visited the website on a weekday or weekend and whether they have made a purchase or not. So there are two variables that are involved in this and then you are also seeing the proportion which is the conversion rate and you can draw a conclusion that on weekend you have a slightly higher conversion rate as compared to the weekdays. If you go by absolute number you might miss this detail but when you do the hundred percent stacked bar graph you can simply capture this saying that 15% is the conversion rate during the weekdays and 17% is the conversion rate during the weekends.

So if you want to run a promotional campaign it would be advisable to do it on a weekend or vice versa if you want to improve the sales throughout the week to better manage your inventory you can focus on weekday sales. So this gives insight to how to run efficiently the business operations and it's a simple graph that is driving these insights. Now we see another type of plot called box and plot.

It is an improvement over box plot where box plot structure is fixed. Here the data is plotted along with the distribution and the outliers. So you can compare the distribution of the data and where is the data stacked or grouped.

you can easily see these features. For example we are seeing two types of graphs here. One is people who are visiting the information page whether they are buying or not buying versus people who are visiting the administrative page and who are buying and not buying.

So the very fact you can see that the distributions of people who are visiting the administrative pages are purchasing more as compared to people who are visiting the information pages and also you can see the distribution of the data under each category and also the outliers together in the box and plot. However for simplicity purpose you can use the box plot first then you can try to use the box and plot to get even deeper insights. So now we will see another type of box plot or an extension of a box plot called swarm plot.

We move from box to box then to swarm. In swarm you are plotting each and every data point so that you know the exact structure of the data without worrying about the bins number of bins. So you can simply see that in this case blue dot is not purchasing, orange dot is purchasing and these are new visitors and these are returning visitors.

You can understand that page values which represent the number of times you are visiting the website is higher for the new visitor as compared to the returning visitors. Returning visitors who are purchasing given by the orange dots are on an average making less than 50 visits or 20 visits whereas the new visitors who are purchasing are definitely making higher visits as compared to a returning visitor. Therefore you

can take this as an insight or an inference into your the way your website is organised whether it is hard for new people to get the information they want before they make a purchase and the returning visitor who is already accustomed to your website is able to navigate faster and make a purchase.

It might be a reason that you want to redesign your website and make it even more customer friendly. You can think about the user experience aspect of that. So these kind of insights can be brought when you are looking at all the data points.

The same inference using a box plot would be much more difficult to derive compared to a box and plot versus a swarm plot. Depending upon the insight that you want to deliver you have to choose the appropriate visual and not for all purposes you should not use swarm plot. Likewise you should not use box plot for everything.

Now we go to multivariate trend analysis understanding the relationship between variables. If you recall from earlier sessions the common way to explore relationship is through scatter plot. You just plot the two variables on x and y axis you get a distribution of the data in the coordinate plane and you can pick a trend line and see whether they are positively correlated or negatively correlated.

For example positive correlation is given here negative correlation you have with one variable is increasing the other variable tends to fall. Then there is no correlation in the sense you can't determine whether there is a positive direction relationship or a negative relationship between them. Then you have linear relationship, exponential, parabolic or u-shaped.

Then you have strong correlation, weak correlation, then there is no correlation. So we also need to talk about the correlation strength whether the positive is strongly positive, negative is strongly negative, whether it is weak, whether it is we can't say anything about it neutral or none. So these kind of things can be immediately seen from a scatter plot.

However there is a very big caveat or a caution that one has to place. Many analysts make this pitfall of assuming correlation means causation. Correlation is only a directional relationship.

It says whether there is a positive directionality between the two variables or negative directionality but it doesn't say or confirm whether one variable is causing another. We tend to impose that meaning because we have a correlation diagram supporting. This is a very big fault and as we will see in a couple of classes later how this can be a very big problem in making assumptions and giving business insights based on correlations that are not substantiated either by business logic or by theory.

So whenever there is a correlation that is not explained by common logic you should always figure that there is some other variables or some other set of variables that might be influencing these two variables together called confounding variables which might be driving this relationship. The relationship that you are trying to see on the XY

scatter plot might be because there are other variables which are not plotted here or are not being analysed can cause this relationship to work in a particular fashion. When you isolate those effects the correlation might not be looking the same.

So one has to be always cautious about drawing a conclusion that correlation means causation and also be looking out for confounding variables which might actually affect the correlation pattern. Now we have a multivariate relationship. Again I was mentioning the scatter plot.

You can fit a trend line and see the strength of correlation and how is it moving. So it is a very easy way to look at the relationship between two variables. Extending this you have a joint plot.

It is an interesting case of scatter plot where the distribution of X and Y variables is also plotted onto the axis so that you can see the scatter plot and get an idea about the relation. At the same time you can also have a look at the type of data distribution that you are having. For example in this case the Y variable is supposed to have a centred data with only one peak whereas the X variable has a very long distribution with outliers and it looks somewhat normal in a particular range.

So this kind of insight you can get through a joint plot and you can use them to see whether the correlation of the relationship that you are finding in the scatter plot whether it is representative of the entire data range or it is for a specific range. In that case if you find the data distribution to be little different you can isolate those points and again run the relationship and see whether the correlation is holding true. So having an idea about the underlying data relationship through a joint plot can make your argument or the inference about correlation even more stronger.

We have seen these plots in the earlier classes pair plot and heap map. Pair plot is simply plotting all the variables in the data set together and the correlations of all the variables in the data set together and see how are they related with each other. So for example in this case you are trying to see the purchase and non-purchase decision being indicated by the blue and red dots or blue and orange dots.

Then you have simply plotted them using different set of variables and you can see what is influencing the purchase decision across the variables and what is the correlation. Similarly a heat map is also plotting of the correlation between the two variables that you find in the data set and the diagonal is always 1 because it is run between the same variables. Generally the heat map plots the set of correlations between the variables in the data set.

Correlation can technically extend from minus 1 to plus 1 and the heat map takes either one direction as the dark shade and the other direction as the light shade depends on the choice. So you can make the positive correlation as a strong colour on the negative correlation as a weak colour or vice versa and the centre you will always have the diagonal element being 1 which shows the correlation of the variable with itself which is always 1. So you can simply look at the heat map and see with the

variables that are correlated with each other strongly either in the positive sense or in the negative sense. So by looking at the colour saturation and looking at the variable type you can understand how are they related and make your inferences.

Now we come to understanding multivariate relationship using parallel coordinates plot. Parallel coordinates are an interesting case where you want to study 3-4 variables together and see how are they related with respect to one particular phenomenon in our case whether the customer is purchasing or not purchasing. Now one interesting feature of parallel plot is it allows you to do a simultaneous comparison as opposed to this it also challenging because you have to bring all the variables to a comparable axis otherwise you would be doing a wrong comparison.

Whenever you are doing a parallel coordinate chart you need to understand the axis has to be confirmable or it has to be compatible with the variables that are being discussed. For example when we look at the page values they indicate number of times the customer is visiting on an average before making a purchase. We can see that the clustering is around 7 to 8 so we can say almost all the people both purchasers and non-purchasers are visiting the website almost 8 to 10 times.

As opposed to this when you look at the exit rates and bounce rates there is no much difference between the purchasers and non-purchasers. So therefore with regard to exit rate and bounce rates the fact of buying and non-buying is not being influenced or what we can see is like the average exit rate and bounce rates between the purchasers and the non-purchasers are almost similar whereas the page visits are clearly distinct. Then we go to multivariate analysis comparison of trends.

Trends means looking the changes over a period of time. This is a very important visual that is very important for business context because you are always trying to grow and you need to understand your growth by comparison with yourself and with your competitors and trend changes is a very important chart or a visualisation that will help you in this process. For example we are seeing the bounce rate trend over the months and we are trying to understand how people who have bought and people who have not bought how are their bounce rates varying over the months.

So the bounce rates are generally falling for both the categories till October which is slight uptick for the people who are buying in October. Then afterwards the bounce rates for the people who have bought has fallen but for the people who have not bought has increased. Similarly the exit rates also have a similar function for people who have bought and not bought.

So now from an operation standpoint you can understand whether the bounce rates for people who are not buying and the exit rates have anything to do with the revenue generation. You can do a deep dive analysis. So there is a sudden change in the trend you can go do a root cause analysis of what why is the reason some change has happened.

So first insight you are doing taking a business action but a simple insight is coming from the visual. So this is the importance of choosing the right graph to showcase what is the change in the trend. And typically line graphs are very interesting because they are simple intuitive to indicate the change in the direction of some phenomenon.

Bounce rates are generally falling. Suddenly there is a change for the people who are not buying. So you can address that concern and they might become your customers.

Now we will be learning about the dual axis chart. Dual axis chart has two axis one on the either side of the graph where you are trying to compare the same variable using one axis and another variable on the other axis and try to see how are they related. Here what are we seeing we are seeing the total number of visitors and the total number of visitors who made purchase.

So we are plotting the total number of visitors on one axis that is the y-axis on to the left. Then you have total number of visitors who made purchase which is on the left axis. So you can simply see during the first four months the trend has remained the same.

July, August, September and October almost the proportion of people who are coming to visit your website and the sale percentage has almost remained the same. In September there is a small uptick. In October and November there is a very dramatic increase in the number of people who are visiting your website.

It has increased from around 500 in a month to 3,000 in a month by November and you had almost seen the number of sale conversions go up from 100 to 700. So this is something an insight that is simultaneously drawn by reading the graph on the x-axis on the y-axis on left hand and the y-axis on the right hand. So a dual axis chart will help you to see the trends in two variables in a common graphic but their scales are different.

On one scale you have 500 to 3,000 which is depicting the number of visitors. On the other scale you have 100 to 700 which is depicting the number of people who have bought. So you are able to compare the trends in one variable with the another variable though their axes are different.

So this is a very interesting application you can do in an Excel itself like a combo chart we will see in the experiential learning session and you can draw conclusion that as visitors increase the sale is also increasing.

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